Quantifying Differences in Hydrogen Peroxide and L-lactate Production across Oral Streptococci using an Optimized Reflectometric Assay

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Abstract

Periodontitis affects 47% of adults in the US. This chronic disease is driven by pathogens including *Aggregatibacter actinomycetemcomitans* (*Aa*), which is associated with a severe form of periodontitis. Further, within the oral cavity, interactions among bacteria influence the development and progression of periodontal disease, and there is potential to leverage these interactions to prevent or treat disease. For instance, the oral symbiont *Streptococcus gordonii* produces hydrogen peroxide (H_2O_2), which can either inhibit or promote *Aa* growth and virulence depending on the spatial context. *S. gordonii* also produces L-lactate, which benefits *Aa* as a carbon source. However, individuals harbor diverse communities of oral streptococci, and our previous data demonstrate that *Aa* fitness varies when cocultured with diverse streptococcal strains. Thus, this project aims to determine whether variation in H_2O_2 and L-lactate production across oral streptococci may contribute to these differences in *Aa* fitness using an optimized reflectometric protocol. Streptococcal cultures were grown in chemically defined medium, normalized by optical density, and incubated in fresh medium before metabolites were quantified using a reflectometric assay and standard curves. Current results demonstrate approximately a 6-fold range observed across strains in H_2O_2 and L-lactate production across streptococcal strains. However, preliminary comparisons suggest that variation in these metabolites alone do not fully explain differences in *Aa* coculture fitness, indicating that additional metabolic or ecological factors may influence these interactions. Together, this work helps identify metabolic factors that alter *Aa* fitness and provides insight into how the unique composition of an individual's microbiome alters periodontal disease pathogenesis.

Introduction

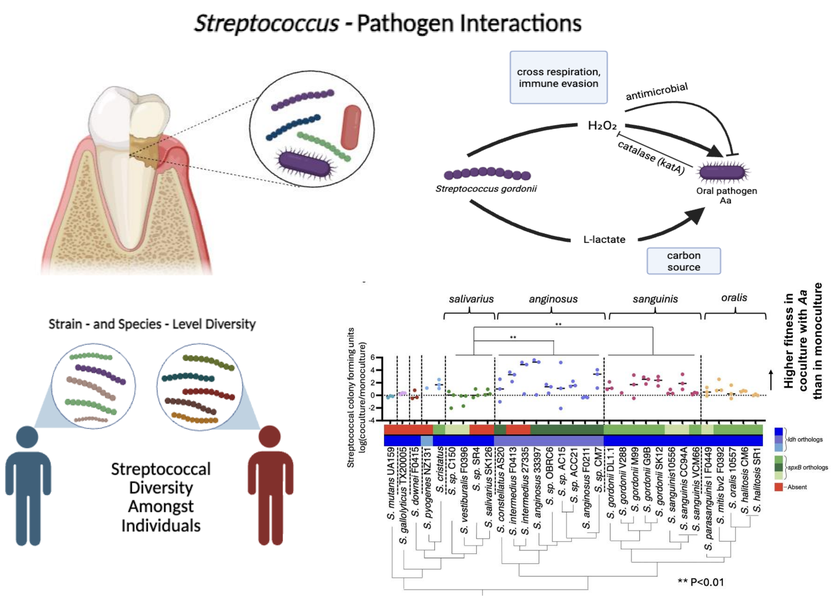


Fig. 1. Rationale for measuring metabolite concentration Oral streptococcal strains exhibit differential fitness during coculture with *Aa* compared with monoculture. Variation in *spxB* (hydrogen perodxide) and *idh* (lactate) orthologs identified by pangenome analysis suggests that strain-specific metabolite production may underlie these fitness differences. Statistical significance was assessed using a Kruskal–Wallis test with Dunn's multiple-comparisons test. Figure made by Grace Heine.

Research Question: Are differences in H_2O_2 and L-lactate production in streptococci associated with strain-specific fitness differences during coculture with *Aa*?

Methods

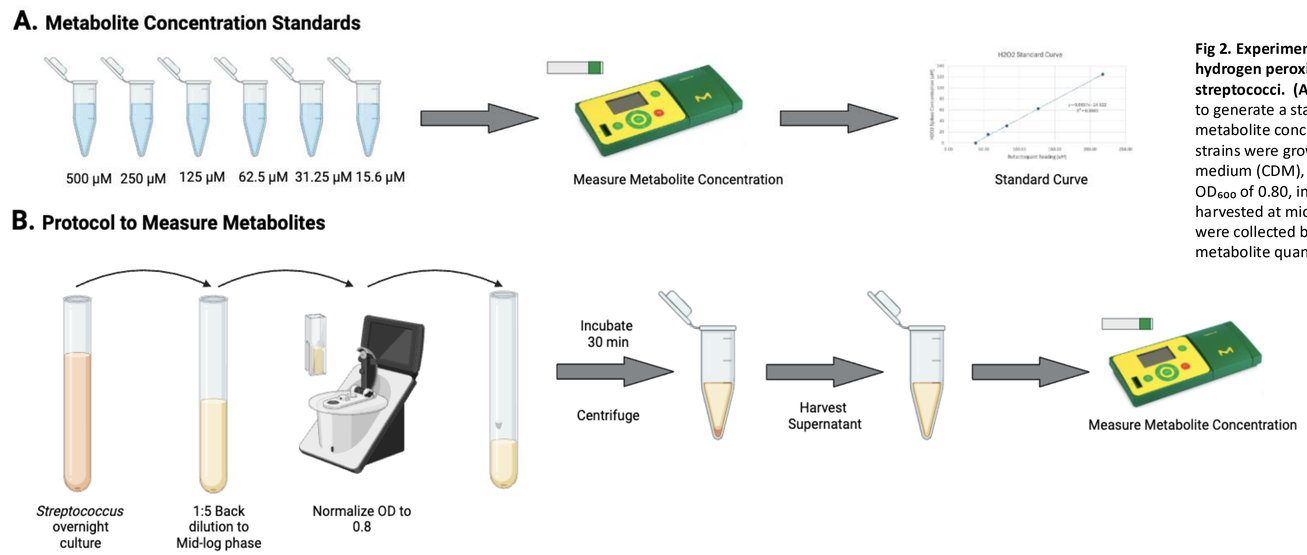


Fig 2. Experimental workflow for quantifying hydrogen peroxide and lactate production by oral streptococci. (A) Serially diluted standards were used to generate a standard curve with an $R^2 > 0.99$ for metabolite concentration. **(B)** Oral streptococcal strains were grown in liquid chemically defined medium (CDM), back-diluted 1:5, normalized to an OD_{600} of 0.80, incubated for 30 minutes and harvested at mid-log phase. Culture supernatants were collected by centrifugation and used for metabolite quantification using a reflectometer.

Results

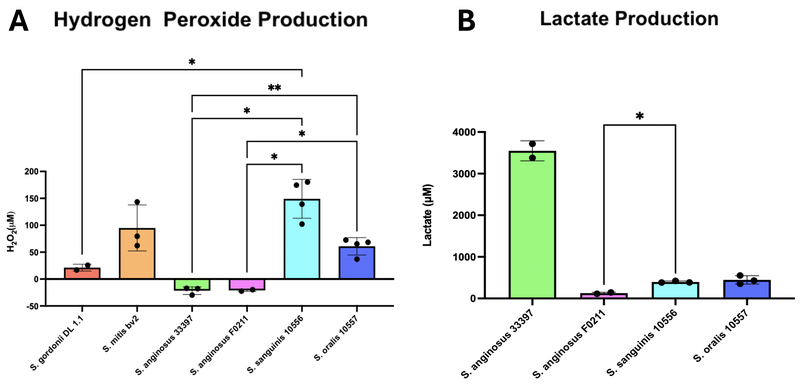
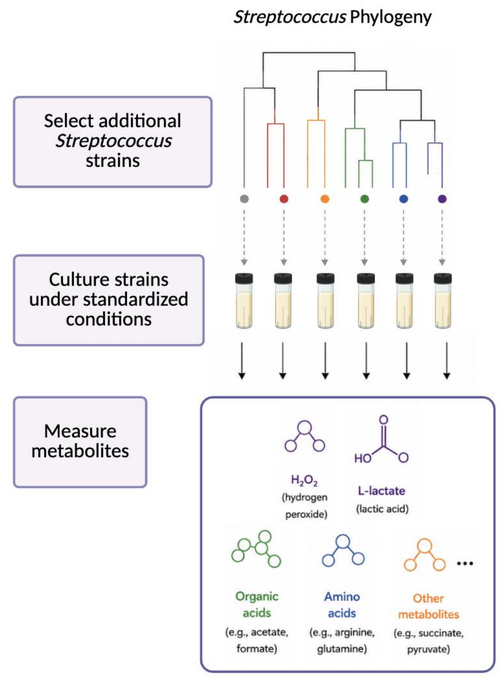


Fig 3. Metabolite production by oral streptococcal strains. Oral streptococcal strains were grown in liquid CDM to mid-log phase, normalized to an OD_{600} of 0.80, and incubated in fresh CDM for 30 minutes. Metabolite concentrations were quantified from culture supernatants. Bars represent the mean ± SD of the biological replicates. Each biological replicate represents the average of technical replicates. **(A)** Hydrogen peroxide production across four oral streptococcal strains. Statistical significance was assessed using ANOVA followed by Tukey's multiple-comparisons test, ****p** < 0.01. **(B)** Lactate production across oral streptococcal strains.

Conclusions

- **Oral streptococcal strains exhibited substantial differences in hydrogen peroxide production, with approximately a 6-fold range observed across strains.**
- **Lactate production was detected in both *S. sanguinis* and *S. oralis*, with comparable concentrations under the conditions tested.**
- **These findings suggest that metabolic output varies among oral streptococci and may contribute to strain-specific ecological interactions.**
- **Future studies will determine how these metabolic differences influence interactions with *Aa*.**

Future Directions



Continue screening additional oral streptococcal strains for H_2O_2 and L-lactate production

Expand metabolite profiling to identify metabolic phenotypes associated with previously observed differences in coculture fitness with *Aa*

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